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Carpe diem: Can daily oil prices improve model-based forecasts of the real price of crude oil?☆

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ABSTRACT

This paper proposes methods to include information from the underlying nominal daily series in model-based forecasts of average real series. We apply these methods to forecasts of the real price of crude oil. Models utilizing information from daily prices yield large forecast improvements and, in some cases, almost halve the forecast error compared to current specifications. We demonstrate for the first time that model-based forecasts of the real price of crude oil can outperform the traditional random walk forecast, that is, the end-of-month no-change forecast, at short forecast horizons.

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1. Introduction

Macroeconomic forecasters often target real series that are aggregated from higher-frequency nominal data. A case in point is the real price of crude oil, which plays a key role in macroeconomic projections and has attracted considerable attention from forecasters (see e.g. Alquist et al., 2013; Baumeister & Kilian, 2012, and the survey by Filippidis et al. 2024). Notably, all existing model-based forecasts of the monthly real price of crude oil rely on monthly averages of daily oil prices, which facilitates incorporating monthly predictor variables and constructing historical price series (Alquist et al., 2013).[2] However, this form of temporal aggregation could also introduce information loss that diminishes forecasting accuracy (see e.g. Kohn, 1982; Lütkepohl, 1986; Marcellino, 1999; Wei, 1978). This issue was recently highlighted by Ellwanger and Snudden (2023a), who demonstrated that the traditional random walk forecast, which is the end-of-month no-change forecast, outperforms all previous model-based forecasts of the real oil price at the one-month forecast horizon.

The goal of this paper is to investigate how daily information can be used to improve model-based forecasts of period-average real series. To do so, we extend existing forecasting approaches dealing with higher-frequency information to a setting where nominal daily data are converted to real monthly average series. We henceforth refer to these methods as *disaggregated* approaches and

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[2] For example, structural models and forecasting approaches using data prior to 1983 typically rely on historical refinery acquisition costs of crude oil, which are only available at the monthly frequency (Alquist et al., 2013; Kilian, 2009).

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apply them to popular forecasting models of the real price of crude oil.

Our main contribution is to compare the effectiveness of disaggregated forecast approaches to forecast approaches that rely on monthly average real data. The proposed extensions provide practical solutions on how to construct forecasts efficiently when the temporal aggregate is expressed in real terms. The analysis quantifies the information loss from daily to monthly aggregation and provides a unique perspective on the forecast gains from temporal disaggregation in practice, which have thus far primarily focused on monthly to quarterly or quarterly to annual aggregation.[3]

Our second contribution is to directly compare the effectiveness of model-based forecasts of the monthly average real price of crude oil to the traditional random walk forecast, which is the end-of-month no-change forecast. In contrast, prior studies have compared model-based forecasts against a monthly average no-change forecast, which can result in spurious predictability (Ellwanger & Snudden, 2023a). Importantly, this marks the first instance of testing forecasts of the monthly average real oil price against the null hypothesis of no predictability.

Our empirical exercises concentrate on the most common models in oil price forecasting, which include recursive forecasts from autoregressive moving average (ARMA) models, vector autoregressive (VAR) models, and direct forecasts from univariate and multivariate models. For each modeling class, we propose practical techniques to integrate disaggregated approaches for period-average real forecasts. Together, these exercises provide a comprehensive comparison of the information loss from temporal aggregation across forecast approaches.

The first disaggregated approach that we consider is the bottom-up (BU) approach. This approach consists of forecasting the underlying high-frequency series and subsequently averaging the forecasts to derive lower-frequency predictions. Theoretical results for ARMA processes show that the BU approach is preferable to models estimated with average data (Kohn, 1982; Tiao, 1972), even under parameter uncertainty (Lütkepohl, 1986). However, it remains an open question how this approach performs in practice for aggregation from daily to monthly variables and for the real price of crude oil in particular. In implementing the BU approach, we propose solutions for how to deal with the aggregation from nominal to real variables, which is a setting frequently encountered in macroeconomic forecasting.

The second disaggregated method is period-end point sampling (PEPS) introduced by Ellwanger and Snudden (2023c). PEPS consists of estimating models with end-of-period data and subsequently using point forecasts as predictions for period averages. By employing end-of-period sampling during estimation, PEPS mitigates information loss caused by temporal aggregation. Moreover, this approach allows models to be estimated at the same frequency as the forecast target and is easy to implement in existing lower-frequency forecasting frameworks.

[3] See, for example, Abraham (1982), Athanasopoulos et al. (2011), Lütkepohl (1986), Zellner and Montmarquette (1971).

The third disaggregated method is mixed-data sampling (MIDAS), which has become a popular approach to deal with higher-frequency predictor variables (see Andreou et al., 2010; Ghysels et al., 2007). Unlike previous MIDAS applications in crude oil price forecasting, which utilize high-frequency financial variables as predictors of the monthly real price of oil (Baumeister et al., 2015; Degiannakis & Filis, 2018; Zhang & Wang, 2019), we employ this framework to forecast monthly oil prices with the underlying daily prices. To our knowledge, this is the first time the MIDAS approach is used for univariate forecasts of lower-frequency data.

Our main finding is that regardless of the approach employed, forecasting models that utilize information from daily prices substantially outperform conventional models estimated with monthly averages, especially at short forecast horizons. The disaggregated methods offer robust improvements in forecast accuracy and can reduce the mean squared forecast error (MSFE) by over 40% at the one-month horizon. Furthermore, models employing daily oil prices can provide higher directional accuracy than traditional forecasting models for forecast horizons up to 24 months ahead. These forecast gains are in line with theoretical predictions (Kohn, 1982; Lütkepohl, 1986; Tiao, 1972) and are substantially larger than those documented in prior empirical investigations, which focused on monthly to quarterly or quarterly to annual aggregation (see e.g. Athanasopoulos et al., 2011; Lütkepohl, 1986; Zellner & Montmarquette, 1971).

Our replications find that existing model-based forecast approaches only outperform the end-of-month no-change at long horizons, and typically only for directional accuracy. In contrast, we demonstrate for the first time that forecasting models of the real price of crude oil can outperform the end-of-month no-change forecast at short forecast horizons when disaggregated techniques are utilized. However, in such cases, the forecast gains are smaller than implied by comparisons with the monthly average no-change forecast. These findings reinforce the idea that the real price of crude oil is more similar to asset prices than previously concluded.

A key insight from our exercises is that much of the benefit of using daily oil prices stems from the information contained in the last daily price in the forecaster's information set. For example, we find that in most cases, MIDAS regressions attribute the entire forecasting weight on the end-of-month price and zero weight on any other daily prices. As a consequence, forecasting models relying on monthly rather than daily data can provide effective forecasts, as long as they incorporate information from end-of-month prices rather than monthly average prices. Although this result is consistent with the well-known fact that the last available instantaneous price contains crucial information about expectations (see e.g. Fama, 1970), this idea is rarely exploited in forecasting applications of average data.

In examining the effectiveness of different forecasting approaches in a setting of temporally aggregated variables, our study also contributes to the broader debate about direct versus recursive forecasting approaches. Existing studies have compared direct and recursive forecast

approaches within a single frequency, suggesting that recursive forecasts often outperform direct forecasts (see e.g. Chevillon, 2007; Ing, 2003; Marcellino et al., 2006). Our findings reveal that while recursive forecasts typically outperform direct forecasts when estimated with monthly average data, they perform similarly when estimated with temporally disaggregated data.

Although the idea to utilize the underlying daily oil price data in forecasts of the monthly real price of oil is new, our results are consistent with several distinct pieces of evidence provided in earlier studies. For instance, Ellwanger and Snudden (2023b) demonstrated that using end-of-period oil futures prices rather than period-average oil futures prices leads to significant improvements in futures-based forecasts of the real oil price. Baumeister and Kilian (2014) showed that aggregating forecasts of monthly average oil prices to quarterly averages is preferable to computing forecasts with quarterly models, while Baumeister and Kilian (2012) showed that forecasting gains from data revisions primarily arise from improved information on crude oil prices. Our study expands on these findings by showing that integrating information from daily prices into model-based forecasts of the monthly real price of crude oil can substantially enhance their accuracy compared to original specifications.

2. Method

The focus of our study is real-time forecasts of the level of the monthly average real price of crude oil, which is the standard approach in the literature (see e.g. Alquist et al., 2013; Baumeister & Kilian, 2012, and the surveys of Ellwanger and Snudden 2023a;Filippidiset al. 2024).

For Brent and WTI, the average monthly nominal price is the simple average of the daily closing prices, $\bar{S}_t = \frac{1}{n}\sum_{i=1}^{n} S_{t,i}$, where $S_{t,i}$ stands for the nominal spot price at the closing of day i in month t, and where n is the number of business-calendar days within a month.[4] Nominal prices are converted to real prices using the monthly seasonally adjusted U.S. consumer price index, denoted by CPI_t. The conventional monthly average real price of crude oil can be expressed as the monthly average of daily real prices, $R_{t,i}$:

$$\bar{R}_t = \frac{\frac{1}{n}\sum_{i=1}^{n} S_{t,i}}{CPI_t} = \frac{1}{n}\sum_{i=1}^{n}\frac{S_{t,i}}{CPI_t} = \frac{1}{n}\sum_{i=1}^{n} R_{t,i}.$$

Following the convention in the literature, we assume that the forecaster uses the available information at the end of month t to form their prediction of the monthly average real price in the h months ahead. The standard approach for forecasts of the real price of crude oil is to ignore daily oil prices and to forecast the h-step-ahead monthly average price, $\bar{R}_{t+h}$, using the history of monthly average oil prices up to $\bar{R}_t$. However, even when the goal is to forecast the monthly average price, incorporating the information contained in higher-frequency price observations could be beneficial for the accuracy of forecasts.

[4] A different oil price series that has been widely used in forecasting applications is the monthly U.S. refiners' acquisition costs of crude oil. We discuss the treatment of this series in Section 2.2.

For example, if the underlying data are generated by an ARMA process, relying on temporally aggregated data will result in a loss of forecast efficiency that increases with the persistence of the underlying data and the number of higher-frequency observations that are aggregated (Kohn, 1982; Tiao, 1972). Moreover, Lütkepohl (1986) demonstrated that utilizing the disaggregated data results in superior forecasts even under parameter uncertainty.[5] Based on these insights, this study examines for the first time how the series of underlying daily oil prices up to $R_{t,n}$ can be used to improve forecasts of the monthly average real price of crude oil.

2.1. Forecast evaluation

We focus on two common forecast evaluation criteria in the literature: the MSFE and the success ratio for directional accuracy. The MSFE ratio for the h-step-ahead forecast, $MSFE_h^{ratio}$, is calculated as the ratio of the MSFE of the model-based forecast to the MSFE of the end-of-month no-change forecast:

$$MSFE_h^{ratio} = \frac{\sum_{t=t_0}^{T}(\bar{R}_{t+h} - \hat{\bar{R}}_{t+h|t})^2}{\sum_{t=t_0}^{T}(\bar{R}_{t+h} - R_{t,n|t})^2}, \quad (1)$$

where $\hat{\bar{R}}_{t+h|t}$ is the model-based forecast conditional on information in time t for the h-step-ahead aggregated observation, $\bar{R}_{t+h}$, and $R_{t,n|t}$ denotes the real-time end-of-month no-change forecast. The evaluation sample begins in t_0 and ends in T. We also report the p-values for the test of equal predictive accuracy relative to the end-of-month no-change forecasts following (Diebold & Mariano, 1995).[6]

The mean directional accuracy is reported using the success ratio, which indicates the proportion of times that the model-based forecasts correctly predict the direction of the change in the monthly average real price. Consistent with previous real-time exercises, the success ratio for forecast horizon h, SR_h, is

$$SR_h = \frac{1}{T}\sum_{t=t_0}^{T}\mathbb{1}[sgn(\bar{R}_{t+h} - R_{t,n}) = sgn(\hat{\bar{R}}_{t+h|t} - R_{t,n|t})], \quad (2)$$

where $\mathbb{1}[\cdot]$ is an indicator function, and $sgn(\cdot)$ is the sign function. The null hypothesis of no directional accuracy (corresponding to a success ratio equal to one half) is tested using (Pesaran & Timmermann, 2009).

The existing practice in the literature has been to report MSFE ratios relative to the conventional monthly average no-change forecast (see e.g. the survey of Filippidis et al., 2024). Doing so replaces the terms $R_{t,n|t}$ and $R_{t,n}$ in Eqs. (1) and (2) with the monthly average

[5] In contrast, Lütkepohl (1984) showed that for contemporaneous aggregation (summation over subcomponents), the gains of using aggregated versus disaggregated data are unclear under parameter uncertainty.

[6] The use of real-time data and the iterative out-of-sample forecasts does not fulfill the assumptions underlying the Diebold–Mariano test (Diebold, 2015). As in extant studies, the p-values are still reported with this caveat in mind.

no-change forecasts, $\bar{R}_{t|t}$ and $\bar{R}_t$, respectively. For reference, we also report all forecasts relative to this naive benchmark.[7]

Comparing model-based forecasts of the monthly average real price against the end-of-month no-change forecast is a separate contribution of this paper. The end-of-month no-change forecast is the only no-change forecast reflecting the null hypothesis that *all* future prices—both daily prices and average prices—are conditionally unpredictable, and therefore corresponds to the traditional random walk forecast used in economics and finance. In contrast, forecast improvements relative to the monthly average no-change forecast are theoretically expected for all (vector) autoregressive integrated moving average representations of the daily data (Marcellino, 1999; McCarthy & Snudden, 2024; Weiss, 1984). Ellwanger and Snudden (2023a) showed that for the real price of crude oil, the end-of-month no-change forecast is significantly more accurate than the monthly average no-change forecasts for up to one year ahead.[8] As such, comparisons with the end-of-month no-change forecast are necessary to provide evidence for the practical usefulness of specific model-based forecast approaches and for the predictability of the monthly average real price of crude oil.

There are two additional reasons to directly report MSFEs and success ratios relative to the end-of-period no-change forecast. First, benchmarking against the end-of-month no-change forecast allows for formal testing against the random walk hypothesis. Second, unlike the MSFE ratios, the success ratios used to summarize directional accuracy are non-transitive with respect to the benchmark forecast. This means that the best performing directional accuracy forecast relative to the monthly average may not be the best directional forecast against the end-of-month no-change forecast.

The series of real end-of-period observations, $R_{t,n}$, is constructed using $R_{t,n} = S_{t,n}/CPI_t$, and the real-time equivalent $R_{t,n|t} = S_{t,n}/CPI_{t|t}$. The measure of real end-of-month prices is consistent with the common practice in other forecast applications, such as bilateral exchange rates (Meese & Rogoff, 1983) and primary commodity prices (West & Wong, 2014). Throughout the paper, significant improvements over the end-of-month no-change forecast are indicated in bold. Corresponding tables reporting results relative to the end-of-month no-change forecast are provided in Appendix A2.

2.2. Disaggregated data

We study model-based forecasts of the three oil price benchmarks used in the literature: West Texas Intermediate (WTI), Brent, and U.S. refiner acquisition cost of crude oil, imported (RAC). Crude oil prices are obtained from the U.S. Energy Information Administration (EIA). The monthly average and daily prices for WTI and Brent are available in real time. The price deflator is the seasonally adjusted U.S. consumer price index obtained from the FRASER database of the Federal Reserve Bank of St. Louis and the real-time database of the Philadelphia Federal Reserve.

Unlike WTI and Brent, the RAC is a survey of the monthly average acquisition costs and is only observed at the monthly frequency. Implementing forecast approaches relying on disaggregated data for the RAC is complicated by the fact that daily RAC data do not exist. For our empirical analysis, we impute daily RAC observations by applying the growth rate of the daily WTI oil price over the monthly average WTI oil price to the monthly RAC price: $S^{RAC}_{t,i} = \bar{S}^{RAC}_t \cdot (S^{WTI}_{t,i}/\bar{S}^{WTI}_t)$, where $S_{t,i}$ refers to the nominal closing price on the day i and the superscripts RAC and WTI denote the nominal RAC series and the nominal WTI prices, respectively.[9] The idea of imputing RAC observations with WTI prices has ample precedent in the oil price forecasting literature. For example, Baumeister and Kilian (2012) nowcasted the average nominal RAC with the growth rate of the WTI price and used WTI futures prices to construct futures-based forecasts for RAC. The tight empirical correlation between WTI prices and the RAC implies that these nowcasts and imputations yield good results in practice.

Our exercises also replicate and extend forecasts from the Kilian and Murphy (2014) VAR model, which includes the real price of crude oil, the growth rate of global crude oil production, a proxy for the change in global crude oil inventories, and an indicator of real economic activity. An additional contribution of this paper is to extend the real-time data of Baumeister and Kilian (2012) using historical data vintages from the EIA's Monthly Energy Review and Short-Term Energy Outlook. Unlike earlier updates of this dataset provided by Garratt et al. (2019), we were able to collect all monthly vintages after 2010M12, which means that this is the first update to include all revisions for the entire history of each vintage.[10] Real-time data vintages start in 1991M1 and contain historical data from 1973M1 onward. Real-time data on U.S. crude oil inventories, U.S. petroleum inventories, and OECD petroleum inventories are obtained from historical releases in the EIA's Monthly Energy Review or the International Data Browser. The real-time version of Kilian (2009)'s real economic activity index is computed using the corrected formula (Kilian, 2019).

All nowcasts follow Baumeister and Kilian (2012). For example, the monthly average RAC is nowcasted using the month-over-month growth rate of the monthly WTI series. Any missing real-time observations for the consumer price index are nowcasted using the average historical growth rate.

The series of daily and monthly WTI and Brent prices provided by the EIA begin in 1986M1 and 1987M5, respectively, and the estimation of models at the daily

[7] See Section 3.1 and Appendix A2.1.

[8] As shown in Table A5 of the online appendix, this also holds for the sample considered in this paper.

[9] Very similar forecasts are obtained by imputing end-of-month RAC prices from Brent rather than WTI prices.

[10] The data appendix contains detailed descriptions of the real-time vintages, nowcasts, and data sources. This real-time database is updated monthly and available publicly on the author's website for download along with further documentation.

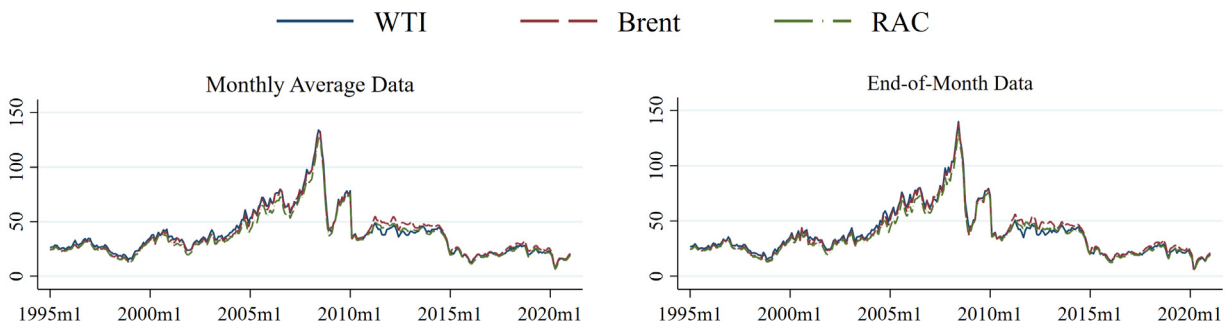


Fig. 1. Real price of crude oil.
Note: Monthly average and end-of-month real prices of crude oil, 1995M1–2021M1, expressed in 2010 USD. 'RAC' is the U.S. refiner acquisition cost of imported crude oil.

frequency begins on those dates. For forecasting models estimated at the monthly frequency, including models estimated via PEPS, prices are backcasted to 1973M1. Specifically, the monthly average and end-of-month prices for WTI and Brent are backcasted to 1983M4 using the growth rate of the monthly average and end-of-month price of the front WTI futures contract, respectively. Then, following standard practice, all monthly average prices before 1983M4 are backcasted using the growth rate of the monthly average RAC. For the backcast of end-of-month prices before 1984M4, we apply the same growth rate to the last available end-of-month observation rather than the last observed monthly average observation. None of these backcasting choices are crucial for our results, and our main findings are robust to estimating models with data starting in 1983 or 1986.

Fig. 1 plots the monthly average and end-of-month real price of crude oil series. Summary statistics are provided in the data appendix. All crude oil series closely track each other. The real prices of crude oil increased during much of the mid-2000s and experienced large movements around the period of the global financial crisis.

3. Model-based forecasts

We focus our analysis on the most common types of model-based forecasts in the literature (see e.g. the survey by Filippidis et al. (2024)). Specifically, we examine disaggregated forecasting approaches for recursive forecasts from univariate ARMA models and from VARs. We also examine direct univariate forecasts from mixed-data sampling (MIDAS) and linear regression models (LRMs) with exogenous predictor variables. The selection of models comprises univariate and multivariate methods for direct and recursive forecasts. Hence, our exercises provide a comprehensive overview of the usefulness of disaggregated approaches in different settings encountered in applied forecasting. All forecasts are evaluated out-of-sample with an expanding window beginning in 1992M1, and the evaluation window ends in 2021M1.

3.1. ARMA forecasts

The most common model-based forecasts for crude oil prices are recursive forecasts from ARMA models. These univariate time-series models commonly serve as an alternative benchmark in forecast exercises and tend to be more accurate predictors of the real price of crude oil than the conventional average-price no-change forecast (Baumeister & Kilian, 2012; Funk, 2018; Snudden, 2018).

The convention in the literature is to estimate ARMA models for real oil prices in log levels. Let the log level of the real price of oil be $\bar{r}_t = \ln(R_t)$, where ln is the natural log function. The ARMA(p, q) with p autoregressive parameters, q moving-average coefficients, and innovations $\check{\epsilon}_t$ is given by

$$\check{a}(L)\bar{r}_t = \check{c} + \check{b}(L)\check{\epsilon}_t \quad \forall t. \tag{3}$$

where $\check{c}$ is a constant and L is the lag operator, such that $Ly_t = y_{t-1}$, $\check{b}(L) = (1 + \check{\alpha}_1 L + \cdots + \check{\alpha}_q L^q)$, and $a(L) = (1 - \check{\rho}_1 L - \cdots - \check{\rho}_p L^p)$. The estimated parameters are used to construct recursive model-based forecasts, $\check{\bar{r}}_{t+h|t}$, which are converted back into real prices in levels, $\check{R}_{t+h|t} = \exp\left(\check{\bar{r}}_{t+h|t}\right)$, where exp is the exponential function.

One advantage of using monthly average data is that the prices of WTI and Brent, which are only available since the early 1980s, can be backcasted to 1973 using the RAC.[11] This is useful because backcasting all series to 1973 provides a better estimate of the long-run mean, which could improve the forecast accuracy at longer horizons. However, this advantage has to be weighed against the efficiency loss from using aggregated data. Therefore, it is an open empirical question of which method is more effective for forecasts of the real price of crude oil.

The first disaggregated approach that we consider for ARMA models is the bottom-up (BU) approach. Empirical evidence suggests that the BU approach improves forecast efficiency in some but not all cases (Abraham, 1982; Athanasopoulos et al., 2011; Lütkepohl, 1986; Zellner & Montmarquette, 1971). However, existing studies have only considered aggregation from monthly to quarterly frequency or from quarterly to annual frequency. To the best of our knowledge, this paper is the first to consider aggregation from daily to monthly frequency. The information loss generally increases when aggregation occurs

[11] The RAC series itself is imputed for the year 1973 following Mork (1989).

over more observations (Tiao, 1972). Moreover, most of the information loss occurs from aggregation over the first few observations. The daily to monthly aggregation moves from no aggregation, $n = 1$, to a substantive aggregation $n = 21$. In contrast, monthly to quarterly aggregation starts from data already aggregated, $n = 21$, to a higher degree of aggregation, $n = 63$. Thus, the information loss from daily to monthly aggregation is expected to be larger than monthly to quarterly aggregation.

One potential complication with constructing BU forecasts for monthly average real data is that CPI is only available at the monthly frequency. This may explain why the BU approach has been overlooked in applications to real macroeconomic variables that are aggregated from a daily frequency (see e.g. Box et al., 2015). For the baseline results, we construct forecasts of the daily nominal price of crude oil. We then average the daily forecasts to the monthly frequency and deflate the monthly average forecast using the CPI forecast. This approach mirrors the current practice of deflating forecasts of nominal monthly average oil prices into real terms using CPI forecasts in futures-based forecasts (Baumeister & Kilian, 2012; Ellwanger & Snudden, 2023b).

Since the nominal daily price is non-stationary, we estimate the model using the growth rate of nominal daily prices, $g_{t,i} = S_{t,i}/S_{t,i-1}$. The ARMA$(p, q)$ model for the growth rate of daily oil prices is given by

$$\hat{a}(L)g_{t,i} = \hat{c} + \hat{b}(L)\hat{\epsilon}_{t,i} \quad \forall i, t, \tag{4}$$

where $\hat{a}(L)$ and $\hat{b}(L)$ are the standard lag polynomials associated with the daily autoregressive and moving average parts, respectively. Estimated parameters are used to construct recursive model-based forecasts of the growth rate of daily nominal crude oil prices, $\hat{g}_{t,n+k|t}$, where k is the forecast horizon in days. The forecasts for the level of the nominal price on day i of month $t + h$, given month t information, are the model-implied cumulative net growth rate between day t, n and day $t+h$, i, denoted by $\hat{g}_{t+h,i|t}$:

$$\hat{S}_{t+h,i|t} = (1 + \hat{g}_{t+h,i|t}) \cdot S_{t,n}. \tag{5}$$

Then, nominal daily forecasts are averaged to the monthly frequency and converted into forecasts of real prices by deflating the monthly nominal forecasts by $\hat{CPI}_{t+h|t}$, the expected CPI deflator:

$$\hat{\tilde{R}}_{t+h|t} = \frac{\frac{1}{n}\sum_{i=1}^{n}\hat{S}_{t+h,i|t}}{E_{t,n}[\hat{CPI}_{t+h|t}]}, \quad \forall h. \tag{6}$$

We use the standard practice of computing CPI forecasts to discount nominal prices. Following (Baumeister & Kilian, 2012), the CPI forecast is constructed by expanding the (nowcasted) current CPI observations with the average historical rate of inflation since 1986M7.

The second disaggregated approach for ARMA models is period-end point sampling (PEPS). This method estimates the model using point-in-time sampled end-of-period data and uses the point forecast to construct a forecast of the average. Ellwanger and Snudden (2023c) documents that the forecast accuracy of the PEPS approach can be both theoretically and empirically identical to that of the BU approach. An important advantage is that the use of point-sampled data avoids the information loss that occurs from estimating models with monthly average data (Kohn, 1982) and allows the model to continue to be estimated at the monthly frequency. Herein, we use the end-of-month forecast as the forecast of the period average, which is a special case. This is suitable for crude oil, since the properties of the daily closing price data are such that forecasts of the end-of-month and monthly average naturally coincide. Moreover, ex post weighted averaging of the end-of-period forecasts was found to result in nearly identical forecast gains, and so are not reported for brevity. This also provides the advantage that it allows us to quantify whether existing end-of-month point forecasts in the literature will be informative priors for forecasts of period averages.

Implementing the PEPS approach is straightforward, as it merely requires replacing the time series of monthly average real prices with the time series of real end-of-period observations, $R_{t,n}$, during the model estimation. The use of end-of-month point-in-time sampling is desirable, as the last observed price reflects the latest information available to the crude oil market participants and the forecaster.

For the PEPS approach, the model estimation closely resembles that of the traditional approach, which relies on monthly average data. Specifically, the ARMA(p, q) model is estimated at the monthly frequency using a time series of end-of-month prices in log-real levels and expressed as

$$\tilde{a}(L)r_{t,n} = \tilde{c} + \tilde{b}(L)\tilde{\epsilon}_t \quad \forall t. \tag{7}$$

Again, the estimates are used to construct recursive model-based forecasts of the end-of-month real price of crude oil $\tilde{r}_{t+h,n|t}$, which are converted back into real prices in levels, $\tilde{R}_{t+h,n|t} = \exp\left(\tilde{r}_{t+h,n|t}\right)$. Then the end-of-month forecasts are used as forecasts of the corresponding monthly average value, $\tilde{\tilde{R}}_{t+h|t} = \tilde{R}_{t+h,n|t}$.

Regarding lag length, previous studies advocate the use of 12 or 24 lags when estimating autoregressive models to predict real crude oil prices (Baumeister & Kilian, 2012). However, it is well known that temporal aggregation will always result in the introduction of moving average terms, even when the data generating process of the daily data is given by an AR process (Telser, 1967; Weiss, 1984). The in-sample analysis of both real and nominal daily crude oil prices suggests an AR model with one or two coefficients when using an information criterion. This is theoretically consistent with an ARMA representation of the monthly average data and an AR(1) or AR(2) representation for the point sampled data used in the PEPS approach. For completeness, we therefore examine the forecast performance from the standard AR(12) model and also include the forecasts from the AR(2) and ARMA(1,1) models. Qualitatively, our subsequent results are not affected by alternative lag specifications considered in the literature. Instead, the information loss from aggregation turns out to be of first-order importance for the model's forecast performance relative to the choice of model parameterization.

Table 1
Real-time ARMA forecasts of the real price of WTI crude oil, versus the monthly average no-change forecast.

Method	Monthly average prices			Bottom-up			Period-end-point sampling		
Model	AR(12)	AR(2)	ARMA(1,1)	AR(12)	AR(2)	ARMA(1,1)	AR(12)	AR(2)	ARMA(1,1)
					MSFE ratio				
1	0.92 (0.207)	0.90 (0.138)	0.91 (0.130)	**0.61** (0.000)	**0.60** (0.000)	**0.60** (0.000)	**0.58** (0.002)	**0.56** (0.001)	**0.56** (0.001)
3	0.96 (0.350)	0.93 (0.275)	0.94 (0.227)	**0.93** (0.021)	**0.92** (0.018)	**0.93** (0.019)	0.87 (0.126)	0.84 (0.113)	**0.84** (0.088)
6	0.99 (0.459)	0.94 (0.334)	0.95 (0.287)	1.04 (0.723)	1.03 (0.666)	1.03 (0.704)	0.93 (0.271)	0.89 (0.215)	0.89 (0.180)
12	1.00 (0.492)	0.94 (0.354)	0.93 (0.265)	1.16 (0.908)	1.14 (0.888)	1.15 (0.910)	0.96 (0.362)	0.90 (0.250)	0.89 (0.190)
24	1.00 (0.499)	0.97 (0.441)	0.92 (0.302)	1.48 (0.989)	1.42 (0.987)	1.45 (0.990)	0.99 (0.470)	0.95 (0.385)	0.92 (0.308)
					Success ratio				
1	0.50 (0.561)	0.52 (0.294)	**0.54** (0.066)	**0.72** (0.000)	**0.72** (0.000)	**0.72** (0.000)	**0.72** (0.000)	**0.71** (0.000)	**0.72** (0.000)
3	0.49 (0.683)	0.50 (0.599)	**0.53** (0.088)	**0.61** (0.000)	**0.61** (0.000)	**0.60** (0.002)	**0.57** (0.008)	**0.58** (0.002)	**0.59** (0.000)
6	0.50 (0.594)	0.50 (0.519)	0.50 (0.403)	**0.57** (0.042)	**0.56** (0.068)	0.55 (0.124)	0.54 (0.191)	0.53 (0.276)	**0.54** (0.096)
12	0.55 (0.215)	0.54 (0.263)	0.54 (0.155)	0.55 (0.143)	0.55 (0.142)	0.53 (0.433)	0.55 (0.196)	0.54 (0.261)	**0.56** (0.064)
24	**0.59** (0.059)	**0.59** (0.059)	0.56 (0.111)	0.51 (0.958)	0.51 (0.718)	0.49 (0.965)	**0.59** (0.060)	**0.61** (0.035)	**0.56** (0.086)

Note: Real-time, out-of-sample forecasts for the monthly average real price of WTI crude oil, 1992M1–2021M1, versus the monthly average no-change forecast. 'Bottom-up' is the ex post averaged forecast of daily prices; 'Period-end-point sampling' uses the end-of-month forecast (Ellwanger & Snudden, 2023c). Brackets report the p-values for the serial dependence robust tests of Pesaran and Timmermann (2009) for the null of no directional accuracy and of Diebold and Mariano (1995) for equal MSFEs. Boldface values indicate statistically significant improvements at the 10% significance level.

To provide a reference point that is consistent with the existing literature, we begin by comparing the model-based forecasts against monthly average price no-change forecasts. The first three columns of Table 1 present the results for the models estimated with the conventional monthly average data. For most models and forecast horizons considered, the model-based forecasts outperform the average price no-change forecasts. Interestingly, in contrast to existing recommendations, the model estimated with two autoregressive parameters outperforms the models estimated with 12 parameters in terms of the MSFE for all potential ending points of the evaluation sample since 2008 at all forecast horizons. For all models, the gains relative to the monthly average no-change forecast appear relatively modest (below 10%) and only constitute significant improvements at the 10% level for directional accuracy at certain forecast horizons.

The corresponding results for the BU approach, displayed in columns 4–6 of Table 1, present a very different pattern. At the one-month horizon, all three models yield about 40% improvements in the MSFE and a directional accuracy of 72%. The magnitude of the improvements at the one-month horizon is remarkable, as the forecasts are more accurate than any other model-based forecast previously reported in the literature. However, the performance of the BU forecasts deteriorates quickly at longer forecast horizons.

The forecasts from the PEPS approach are evaluated in the final three columns of Table 1. For all horizons and models, the forecasts outperform the conventional monthly average no-change benchmark. At one month ahead, the PEPS approach results in the largest improvements in forecast accuracy of up to 44%. For all three models, forecast criteria, and for all horizons, the models estimated with end-of-period prices provide forecasts of average prices that are at least as good as those of models estimated with average prices.

The deterioration of the BU forecasts at longer horizons is because the model is estimated in growth rates, which loses information about the historical mean. For robustness, we also consider BU forecasts for the level of real daily prices. Following Froot and Ramadorai (2005), we impute a daily CPI by assigning the monthly CPI index to the last day of the month and use linear interpolation to impute a daily CPI measure. The model is then estimated using the log level of the daily real prices, and forecasts of daily real prices are averaged to the monthly frequency. The results, presented in the online appendix, show little difference in short horizon forecast accuracy between the distinct BU approaches. The results also show that applying the BU approach to daily prices in levels instead of growth rates slightly improves longer-horizon forecasts, albeit by less than the improvements obtained under the PEPS approach. This pattern can be explained by the fact that forecasts from models with daily prices only use data beginning in 1986M1, whereas models estimated with monthly average data and PEPS use data that start in 1973M1. The results highlight the importance of using improved information in crude oil prices, while alternative assumptions for the CPI appear to play only a minimal role.

For each oil price series, Fig. 2 reports the evolution of the MSFE and the success ratios for the one-month-ahead forecast from the AR(2) models estimated with average, end-of-month, and daily data. The forecast performance of the BU approach closely tracks that of the end-of-month no-change forecast throughout the forecast evaluation sample, whereas the PEPS forecasts tend to outperform the end-of-month no-change forecast in the latter half of the sample. In contrast, for all series and all potential ending points of the evaluation sample, the models estimated with average prices do not outperform the end-of-month no-change forecast. For every potential ending point of the forecast evaluation period, the disaggregated approaches outperform the models estimates with monthly average data. Moreover, the disaggregated approaches are statistically more accurate than the monthly average no-change forecast at the 5% level for WTI and RAC crude oil prices for all potential ending dates of the forecast evaluation sample between 1997M1 and 2021M1.

Table 2 reports the same model-based forecasts against the end-of-period no-change forecast. The absence of bold values highlights that, even at the 10% significance level, none of the forecasts provide significant MSFE gains over the end-of-month no-change forecast at horizons of less

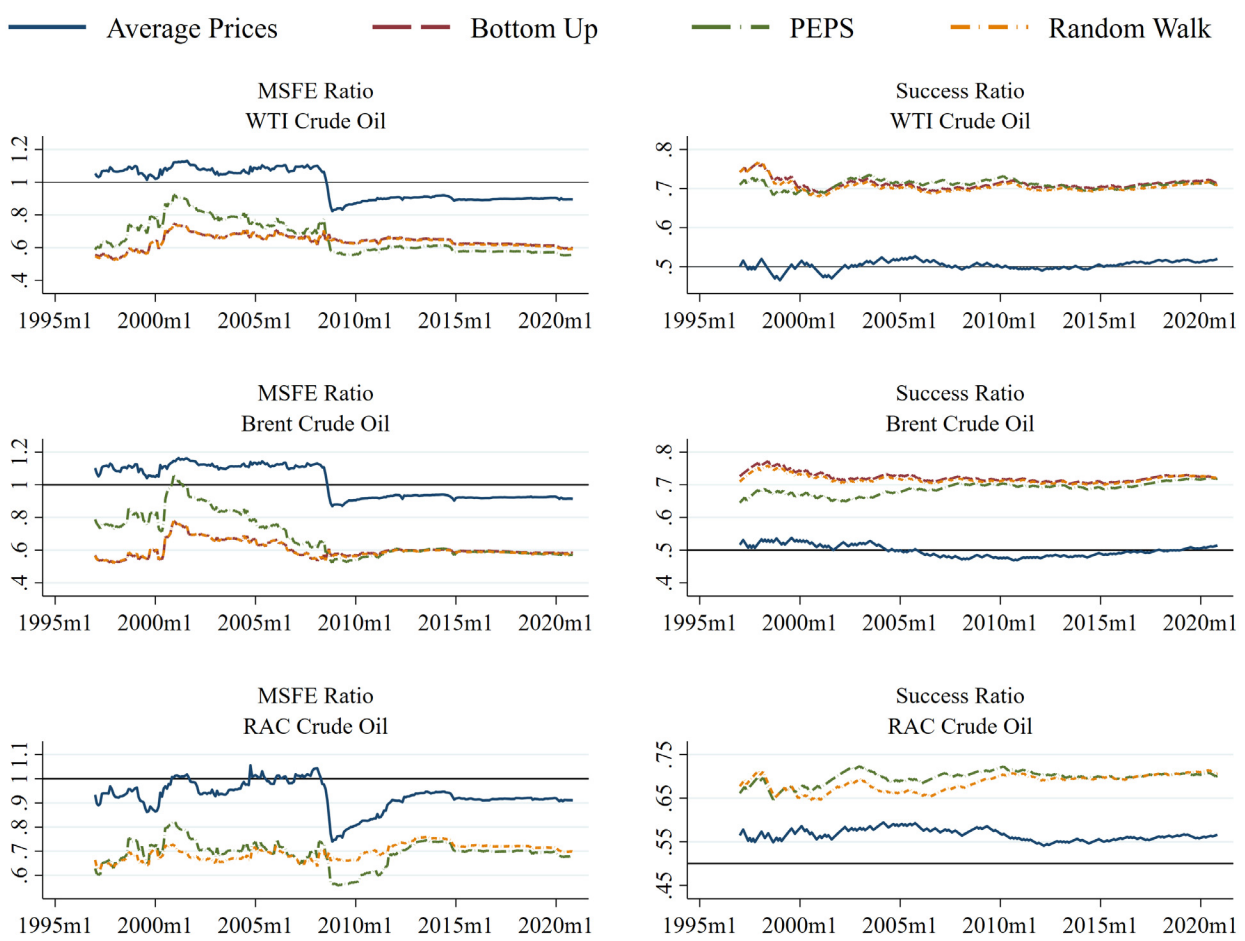


Fig. 2. Evolution of one-month-ahead forecast evaluation criteria for AR(2) models.
Note: One-month-ahead, out-of-sample, real-time forecasts of the monthly average real prices of crude oil, 1992M1–2021M1. All forecasts are compared relative to the period-average no-change. 'Random Walk' refers to the end-of-month no-change forecast. Other forecasts are generated with an AR(2) constructed using monthly 'Average Prices', the 'Bottom Up' approach which uses daily data, or the 'PEPS' approach which uses end-of-month data. The first 60 months are omitted to reduce starting-point effects.

Table 2
Real-time ARMA forecasts of the real price of WTI crude oil, versus the end-of-month no-change forecast.

Method	Monthly average Prices			Bottom-up			Period-end-point sampling		
Model	AR(12)	AR(2)	ARMA(1,1)	AR(12)	AR(2)	ARMA(1,1)	AR(12)	AR(2)	ARMA(1,1)
					MSFE ratio				
1	1.56 (1.000)	1.51 (1.000)	1.54 (1.000)	1.03 (0.878)	1.01 (0.874)	1.02 (0.901)	0.99 (0.439)	0.94 (0.250)	0.94 (0.248)
3	1.09 (0.883)	1.05 (0.731)	1.06 (0.873)	1.05 (0.901)	1.04 (0.885)	1.05 (0.909)	0.98 (0.402)	0.95 (0.307)	0.95 (0.274)
6	1.04 (0.666)	0.99 (0.485)	1.00 (0.495)	1.10 (0.907)	1.08 (0.895)	1.09 (0.909)	0.99 (0.445)	0.94 (0.326)	0.94 (0.300)
12	1.04 (0.612)	0.98 (0.457)	0.97 (0.396)	1.21 (0.961)	1.19 (0.955)	1.20 (0.965)	1.00 (0.493)	0.94 (0.350)	0.92 (0.293)
24	1.01 (0.530)	0.99 (0.471)	0.94 (0.337)	1.50 (0.992)	1.44 (0.991)	1.47 (0.993)	1.00 (0.501)	0.96 (0.416)	0.93 (0.340)
					Success ratio				
1	0.45 (0.986)	0.47 (0.912)	0.47 (0.895)	0.51 (0.431)	0.48 (0.732)	0.50 (0.430)	**0.55** (0.043)	**0.55** (0.034)	**0.54** (0.063)
3	0.50 (0.601)	0.48 (0.791)	0.50 (0.353)	0.53 (0.629)	0.54 (0.393)	0.52 (0.682)	0.48 (0.753)	0.48 (0.782)	0.49 (0.582)
6	0.47 (0.793)	0.48 (0.785)	0.49 (0.495)	0.55 (0.547)	0.56 (0.301)	0.54 (0.486)	0.52 (0.425)	0.49 (0.678)	0.48 (0.649)
12	0.52 (0.378)	0.52 (0.411)	0.50 (0.486)	0.50 (1.000)	0.47 (0.955)	0.46 (0.982)	0.55 (0.173)	0.52 (0.401)	0.54 (0.200)
24	**0.60** (0.048)	**0.61** (0.033)	0.55 (0.161)	0.48 (1.000)	0.44 (0.999)	0.42 (1.000)	**0.60** (0.044)	**0.60** (0.042)	**0.58** (0.060)

Note: Real-time, out-of-sample forecasts for the monthly average real price of WTI crude oil, 1992M1–2021M1, versus the end-of-month no-change forecast. 'Bottom-up' is the ex post averaged forecast of daily prices; 'Period-end-point sampling' uses the end-of-month forecast (Ellwanger & Snudden, 2023c). Brackets report the p-values for the serial dependence robust tests of Pesaran and Timmermann (2009) for the null of no directional accuracy and of Diebold and Mariano (1995) for equal MSFEs relative to the end-of-month (traditional random walk) no-change forecast. Boldface values indicate statistically significant improvements at the 10% significance level.

than two years. This demonstrates that the gains relative to the monthly average no-change forecasts are not indicative of gains relative to naive forecasts. Only the PEPS forecasts significantly predict the direction of change in prices at one month ahead relative to the end-of-month no-change forecast.

Moreover, it is evident that the choice of the no-change forecast changes the relative model performance. For example, at the one-month horizon, the BU forecasts perform about as well as the PEPS when both are compared to the monthly average no-change forecast, whereas PEPS outperforms in Table 2. All subsequent tables are reported relative to end-of-month no-change forecast.[12]

Overall, these results indicate that suitably constructed univariate time-series models that exploit disaggregated prices outperform models estimated with monthly average prices. Moreover, models estimated with disaggregated oil prices provide statistically significant improvements in directional accuracy over the end-of-month no-change forecast at the one-month horizon. These findings are consistent with theoretical results indicating that the information gains from using daily rather than monthly average data may be substantial, particularly for short forecast horizons (Wei, 1978). The gains from using daily data in our setting are much greater than the gains from using monthly series instead of quarterly aggregated series (see e.g. Abraham, 1982; Athanasopoulos et al., 2011; Baumeister & Kilian, 2014; Lütkepohl, 1986; Zellner & Montmarquette, 1971).

3.2. Vector autoregressive models

Another recursive forecast method to investigate is the VAR model of the global market for crude oil (Kilian & Murphy, 2014), which was first applied for real-time forecasts by Baumeister and Kilian (2012) and has been widely used and extended in the literature (see e.g. Funk, 2018; Snudden, 2018, among others). Since our objective is to extend forecast methods using daily prices and not to take a stand on the best forecast approach, we focus on the original specification of Baumeister and Kilian (2012) for illustrative purposes, since it remains the benchmark for alternative VAR specifications. The model contains four variables: the monthly average real price of crude oil, the growth rate of global crude oil production, a proxy for the change in global crude oil inventories, and the real economic activity index (REA) as a measure of global economic activity. It is not possible to apply the BU approach to the oil market VAR as the three non-price variables are only observed at the monthly frequency. Mixed frequency techniques have already been explored for this VAR (Baumeister et al., 2015), so we explore the application of the PEPS approach given its suitability, as shown in the previous section. The exercise illustrates how high-frequency information can be incorporated in large multivariate (policy) models intended to forecast period averages, without having to change the frequency at which the model is estimated.

[12] Comparisons against the monthly average no-change forecast can be found in the appendix.

Similar to the ARMA model examined in Section 3.1, the convention in the literature is to estimate models with the log level of the real monthly average price of crude oil. The VAR(p) model with p autoregressive parameters can be expressed as

$$(1 - \mathbf{a}(L))\mathbf{y}_t = \mathbf{e}_t \quad \forall t. \tag{8}$$

where $\mathbf{y}_t$, is a 4×1 vector of the variables, $\mathbf{a}(L)$ is the autoregressive parameter matrix of order p, and $\mathbf{e}_t$ is a 4×1 vector of innovations. The estimated parameters are used to construct recursive model-based forecasts of the log-real price, $\breve{\breve{r}}_{t+h|t}$, which are converted back into real prices in levels using $\breve{\breve{R}}_{t+h|t} = \exp\left(\breve{\breve{r}}_{t+h|t}\right)$.

Applying PEPS to the VAR setting only involves replacing the log level of the monthly average real price of crude oil, $\bar{r}_t$, in the vector $\mathbf{y}_t$, with the log level of the real end-of-month price, $r_{t,n}$. Again, the estimated parameters of this new specification are used to construct recursive forecasts of the end-of-month real price of crude oil $\tilde{r}_{t+h,n|t}$, which are converted back into real prices in levels, $\tilde{R}_{t+h,n|t} = \exp\left(\tilde{r}_{t+h,n|t}\right)$. Then, the end-of-month forecasts are used as forecasts of the corresponding monthly average value, $\tilde{\bar{R}}_{t+h|t} = \tilde{R}_{t+h,n|t}$.

The estimation follows the real-time forecasts of Baumeister and Kilian (2012). Accordingly, we estimate models with 12 autoregressive lags, although qualitatively similar results are obtained for models with 24 lags. Other than changing average prices to end-of-month prices, no other change is made, and the VAR models are estimated at the monthly frequency.

Table 3 reports the forecast performance of the VAR models for the standard specification, as well as the PEPS extension, relative to the end-of month no-change forecast. For most cases, the VAR models estimated with PEPS outperform the VAR models estimated with monthly average prices. This is particularly true for forecast horizons of up to six months. The MSFE improvements in forecast accuracy at the one-month horizon are 0.52 percentage points, and, as shown in the online appendix, robust over the second half of the forecast evaluation sample. Gains in directional accuracy relative to the end-of-period no change are achieved at the one-, six-, 12-, and 24-month horizons. These results illustrate how the use of end-of-month prices in VARs helps to restore the information loss introduced from averaging, which is particularly relevant for short-horizon forecasts.

3.3. MIDAS forecasts

An alternative disaggregated approach that we examine is MIDAS (Andreou et al., 2010; Ghysels et al., 2007). The MIDAS approach constructs direct forecasts of lower-frequency variables with higher-frequency predictors. Previous applications of the MIDAS model to oil price forecasts have focused on the role of high-frequency financial variables as predictors (Baumeister et al., 2015; Degiannakis & Filis, 2018; Zhang & Wang, 2019). Instead, we explore the relationship between daily and monthly prices of crude oil in a univariate setting. To the best of our knowledge, the MIDAS approach has

Table 3
Forecasts of the real price of crude oil with VAR models, versus the end-of-month no-change forecast.

Estimation	Monthly average prices			Period-end-point sampling		
Horizon	WTI	Brent	RAC	WTI	Brent	RAC
			MSFE ratio			
1	1.74 (1.000)	1.89 (1.000)	1.36 (0.989)	1.22 (0.890)	1.42 (0.993)	1.07 (0.639)
3	1.18 (0.895)	1.25 (0.960)	1.17 (0.823)	1.09 (0.719)	1.22 (0.927)	1.08 (0.659)
6	1.16 (0.950)	1.23 (0.974)	1.20 (0.934)	1.10 (0.863)	1.23 (0.972)	1.15 (0.885)
12	1.21 (0.950)	1.22 (0.949)	1.20 (0.924)	1.18 (0.915)	1.24 (0.948)	1.17 (0.900)
24	1.09 (0.712)	1.09 (0.709)	1.06 (0.644)	1.10 (0.711)	1.11 (0.727)	1.07 (0.653)
			Success ratio			
1	0.45 (0.950)	0.46 (0.920)	**0.55** (0.057)	**0.54** (0.075)	0.53 (0.102)	**0.54** (0.052)
3	0.51 (0.315)	0.52 (0.170)	0.53 (0.127)	0.51 (0.283)	0.53 (0.127)	0.53 (0.166)
6	0.50 (0.423)	0.52 (0.214)	0.52 (0.186)	0.53 (0.233)	0.54 (0.120)	**0.58** (0.012)
12	0.54 (0.101)	**0.56** (0.051)	**0.54** (0.046)	**0.57** (0.035)	**0.59** (0.013)	**0.55** (0.062)
24	**0.57** (0.053)	**0.59** (0.013)	**0.55** (0.065)	**0.56** (0.093)	**0.58** (0.038)	**0.57** (0.045)

Note: Real-time, out-of-sample-forecasts of the real price of crude oil, 1992M1–2021M1, versus the end-of-month no-change forecast. VAR models are estimated with 12 lags. Brackets report the p-values for serial dependence robust tests of Pesaran and Timmermann (2009) for the null of no directional accuracy and of Diebold and Mariano (1995) for equal MSFEs relative to the end-of-month (traditional random walk) no-change forecast. Boldface values indicate statistically significant improvements over the end-of-month no-change forecast at the 10% significance level.

Table 4
Direct forecasts of the real price of crude oil with MIDAS and PEPS, versus the end-of-month no-change forecast.

Method	Monthly average prices			MIDAS			Period-end-point sampling		
Series	WTI	Brent	RAC	WTI	Brent	RAC	WTI	Brent	RAC
					MSFE ratio				
1	1.69 (1.000)	1.73 (1.000)	1.43 (0.999)	0.99 (0.440)	1.02 (0.669)	0.98 (0.227)	0.99 (0.268)	1.00 (0.453)	**0.99** (0.063)
3	1.13 (0.991)	1.11 (0.986)	1.13 (0.987)	0.99 (0.411)	1.01 (0.576)	0.99 (0.331)	1.00 (0.429)	1.01 (0.634)	0.99 (0.318)
6	1.04 (0.810)	1.03 (0.747)	1.04 (0.809)	0.98 (0.344)	1.00 (0.505)	0.98 (0.345)	0.98 (0.344)	1.00 (0.531)	0.98 (0.338)
12	0.98 (0.426)	0.99 (0.458)	1.00 (0.501)	0.94 (0.263)	0.98 (0.382)	0.95 (0.301)	0.94 (0.254)	0.98 (0.365)	0.96 (0.291)
24	0.91 (0.211)	0.92 (0.195)	0.92 (0.206)	0.90 (0.171)	0.91 (0.182)	0.90 (0.152)	0.90 (0.165)	0.91 (0.160)	0.91 (0.144)
					Success ratio				
1	0.45 (0.972)	0.43 (0.998)	0.50 (0.482)	0.52 (0.245)	0.48 (0.733)	0.53 (0.123)	0.52 (0.283)	0.51 (0.477)	**0.55** (0.041)
3	0.47 (0.882)	0.50 (0.525)	0.48 (0.805)	0.51 (0.558)	0.50 (0.557)	0.51 (0.521)	0.50 (0.649)	0.51 (0.462)	0.49 (0.687)
6	0.49 (0.760)	0.52 (0.383)	0.54 (0.202)	0.51 (0.540)	0.54 (0.261)	0.54 (0.219)	0.51 (0.598)	0.54 (0.260)	0.53 (0.320)
12	0.53 (0.345)	0.56 (0.126)	0.55 (0.222)	0.52 (0.421)	0.54 (0.210)	0.54 (0.294)	0.52 (0.426)	0.55 (0.164)	0.55 (0.238)
24	**0.60** (0.039)	**0.63** (0.013)	**0.62** (0.023)	**0.59** (0.053)	**0.63** (0.014)	**0.60** (0.050)	**0.59** (0.054)	**0.63** (0.019)	**0.60** (0.050)

Note: Real-time, out-of-sample-forecasts of the real price of crude oil, 1992M1–2021M1, versus the end-of-month no-change forecast. Brackets report the p-values for serial dependence robust tests of Pesaran and Timmermann (2009) for the null of no directional accuracy and of Diebold and Mariano (1995) for equal MSFEs relative to the end-of-month (traditional random walk) no-change forecast. Boldface values indicate statistically significant improvements over the end-of-month no-change forecast at the 10% significance level.

only been applied to exogenous predictor variables under the assumption that the underlying higher-frequency data of the forecast variable is not observed. Our analysis thus provides a novel application of MIDAS to temporally aggregated, univariate variables.

MIDAS forecasts of the monthly average real price of crude oil are constructed using daily oil prices via

$$\bar{R}_{t+h} = a_h + \beta_h B(L^{0/m}; \theta_h) R_{t,n} + \epsilon_{t+h}, \tag{9}$$

where $\bar{R}_{t+h}$ is the monthly average real price of crude oil, m is the maximum number of daily lags used in estimation, $L^{1/m} R_{t,n} = R_{t,n-1}$ denotes the lag operator at the daily frequency, and the restrictions on the parameters on the daily data are given by a lag distribution $B(L^{0/m}; \theta_h)$, for instance the beta function or the Almon lag.

In this framework, the estimates of the restricted parameter profiles provide the weights given to the different daily prices. We estimate two of the most common restricted parameter specifications: the normalized exponential Almon lag weight function, and the normalized beta lag polynomial, both for alternative lag lengths of up to 20 daily observations.

Interestingly, the full-sample estimates are conclusive that for WTI and RAC, the coefficients on all daily prices except the last daily price are zero. A similar pattern is obtained for Brent, except for the three- and six-month-ahead forecasts, for which there are some small non-zero coefficients on daily observations beyond one lag. The finding is consistent with the results of Section 3.1, suggesting that the last observed daily oil price contains practically all the relevant information for forecasting the monthly average real price of crude oil in a univariate setting.

In the case that the only non-zero weight is on the last observation, Eq. (9) simplifies to the following regression equation:

$$\bar{R}_{t+h} = \hat{\alpha}_h + \hat{\beta}_h R_{t,n} + \hat{\epsilon}_{t+h}. \tag{10}$$

That is, only the last observed daily closing price is used to provide direct forecasts of future average prices. Eq. (10) can be interpreted as a special case of the unrestricted MIDAS (UMIDAS) approach of Foroni et al. (2015, 2019), which implies estimating direct relationships between the underlying high-frequency observations and the target variable instead of estimating a parameterized polynomial function. The UMIDAS specification also has the benefit of highlighting the conceptual differences between the alternative disaggregated approaches. For the remainder

of the section, we therefore work with UMIDAS forecasts of Eq. (10).

Direct forecasts of the real price of oil using real end-of-month prices can also be implemented via PEPS. In this case, the model is estimated with real end-of-month prices as the dependent variable,

$$R_{t+h,n} = \tilde{\alpha}_h + \tilde{\beta}_h R_{t,n} + \tilde{\epsilon}_{t+h}. \quad (11)$$

Then the end-of-month forecast is used as a forecast of the monthly average price, $\tilde{\bar{R}}_{t+h} = \tilde{R}_{t+h,n}$.

Finally, we contrast the MIDAS and PEPS forecasts with direct forecasts estimated on average prices,

$$\bar{R}_{t+h} = \check{\alpha}_h + \check{\beta}_h \bar{R}_t + \check{\epsilon}_{t+h}. \quad (12)$$

For all forecasts, parameters are estimated with an expanding sample window using real-time data, and forecasts are computed out-of-sample.

Table 4 presents the MIDAS and PEPS forecasts along with the direct forecasts constructed from the monthly average prices, and reported relative to the end-of-month no-change forecast. It shows that the MIDAS forecasts (columns 4–6) and the PEPS forecasts (columns 7–9) are very similar for all three oil price series and across all forecast horizons. In terms of MSFE, for all series and forecast horizons, both disaggregated forecasts are at least as accurate as the direct forecasts estimated with average prices (columns 1–3), and at short-horizons outperform them by a substantial margin. These forecast gains are robust across the entire forecast evaluation sample (see Figure A1).

Generally, the forecast gains are very similar to those obtained from the end-of-month no-change forecasts. The bold values indicate that at the 10% significance level, directional accuracy and mean-squared gains relative to the end-of-month no-change forecast are achieved for the one-month-ahead PEPS forecasts for RAC. At the two-year horizon, all forecasts exhibit significant directional accuracy relative to the end-of-month no-change forecasts.

The direct forecasts from both the MIDAS and PEPS applications exhibit the largest gains in accuracy at the two-year horizon for any of the methods explored. Moreover, the forecast gains from using disaggregated oil prices in a univariate MIDAS framework are larger than those documented in previous studies that incorporate high-frequency financial predictors.

A key implication from the exercises presented in this section is that in practice, it makes little difference whether the forecaster uses the real monthly average prices or the real end-of-month price as the target variable in the estimation of forecasts of the monthly average real price. By contrast, using the information contained in disaggregated oil prices, and in particular the last observed oil price in the forecaster's information set, is crucial for constructing accurate forecasts. The simple MIDAS models of this section clearly highlight this point and provide a strong motivation for considering disaggregated observations of the predicted variable in other MIDAS applications involving high-frequency predictors.

The analysis of the MIDAS and BU forecasts also provides evidence of the relative merits of direct and recursive forecasts in the context of temporal disaggregation. Previous comparisons of direct and recursive forecasts assume that variables are observed at the same frequency (see e.g. Chevillon, 2007; Ing, 2003; Marcellino et al., 2006). These studies generally suggest that recursive methods provide better forecasts compared to direct methods. However, the forecasts from the direct MIDAS approach and the recursive BU forecasts tend to be equally accurate at short horizons. In contrast, direct forecast approaches show some advantages at longer horizons. For both direct and recursive approaches, we find comparable and substantial improvements upon the forecasts constructed using averaged data. Again, the findings confirm that accounting for temporal aggregation is of first-order importance compared to other modeling choices.

3.4. Exogenous predictor variables

In this section, we examine the usefulness of daily oil prices in predicting the real price of crude oil in a setting with exogenous predictor variables. The most common way to incorporate exogenous predictors is via direct forecasts from linear regression models (LRMs). Such models are widely used to provide individual forecasts of the real price of crude oil (see e.g. Alquist et al., 2013; Baumeister & Kilian, 2012; Chen, 2014) and are also key components of model averaging exercises (Baumeister & Kilian, 2015; Baumeister et al., 2014; Garratt et al., 2019; Wang et al., 2015). The predictive regressions in previous studies take on the form

$$\check{\bar{R}}_{t+h} = \bar{R}_t \cdot (1 + \check{\alpha}_h + \check{\beta}_h X_t), \quad (13)$$

where X_t is the monthly percent change of predictor variable, and $\check{\alpha}_h$ and $\check{\beta}_h$ are the least-squares estimates.

For illustrative purposes, this section focuses on a predictor variable that has been shown to yield particularly accurate one-month-ahead forecasts of the real price of crude oil, namely the NYSE Arca (AMEX) oil index, a portfolio of oil-sensitive stock prices (Chen, 2014). The predictor variable, X_t, is constructed using the end-of-month AMEX stock index, $p_{t,n}$, in targeted monthly growth rates, i.e. $X_t = p_{t,n}/p_{t-j,n} - 1$, where $j = 1$ when $h = 1$, and $j = h/2$ otherwise, following (Snudden, 2018).

In addition to the standard specification of Eq. (13), we propose a version of the predictive regression where the starting point of the regression is not the monthly average real price, but rather the end-of-month real price. This specification reads

$$\hat{\bar{R}}_{t+h} = R_{t,n} \cdot (1 + \hat{\alpha}_h + \hat{\beta}_h X_t). \quad (14)$$

That is, the monthly average price, $\bar{R}_t$, on the right-hand side of Eq. (14) is replaced with the end-of-month price, $R_{t,n}$. Intuitively, this means that the forecast is started from the end-of-month real price, which is the last non-averaged price available to forecasters.

In addition, we consider the PEPS specification of the LRM, which produces a direct forecast of the end-of-month real price via

$$\tilde{R}_{t+h,n} = R_{t,n} \cdot (1 + \tilde{\alpha}_h + \tilde{\beta}_h X_t). \quad (15)$$

As for the other PEPS specifications, the forecast of the end-of-month price is then used as a forecast of the monthly average price, i.e. $\tilde{\bar{R}}_{t+h} = \tilde{R}_{t+h,n}$.

Table 5
Real-time forecasts of the real price of crude oil using the NYSE Arca Oil Index, versus the end-of-month no-change forecast.

Method	Monthly average prices			End-of-month prices			Period-end-point sampling		
Price	WTI	Brent	RAC	WTI	Brent	RAC	WTI	Brent	RAC
					MSFE ratio				
1	1.36 (1.000)	1.34 (1.000)	1.29 (1.000)	**0.96** (0.074)	**0.95** (0.009)	1.01 (0.629)	**0.95** (0.094)	**0.93** (0.015)	1.01 (0.648)
3	1.05 (0.949)	1.01 (0.669)	1.08 (0.992)	0.97 (0.181)	0.97 (0.148)	1.00 (0.472)	0.97 (0.183)	0.97 (0.160)	1.00 (0.458)
6	1.04 (0.982)	1.01 (0.795)	1.05 (0.994)	0.99 (0.193)	0.99 (0.225)	1.01 (0.698)	0.99 (0.132)	0.99 (0.178)	1.01 (0.678)
12	1.03 (0.961)	1.01 (0.747)	1.06 (0.994)	0.99 (0.222)	0.99 (0.228)	1.02 (0.907)	0.99 (0.246)	0.99 (0.225)	1.02 (0.908)
24	0.97 (0.158)	**0.96** (0.042)	1.09 (0.953)	**0.95** (0.056)	**0.94** (0.027)	1.07 (0.913)	**0.96** (0.063)	**0.94** (0.027)	1.07 (0.930)
					Success ratio				
1	0.46 (0.947)	0.46 (0.955)	0.50 (0.486)	**0.54** (0.056)	**0.58** (0.002)	**0.53** (0.094)	**0.54** (0.056)	**0.58** (0.001)	**0.53** (0.094)
3	0.50 (0.505)	0.50 (0.568)	0.52 (0.251)	0.52 (0.306)	0.52 (0.315)	**0.54** (0.091)	0.52 (0.315)	0.52 (0.315)	**0.54** (0.091)
6	0.51 (0.263)	0.49 (0.657)	0.49 (0.597)	0.54 (0.150)	0.53 (0.223)	0.46 (0.877)	0.52 (0.185)	0.52 (0.245)	0.46 (0.899)
12	0.51 (0.298)	0.51 (0.339)	0.47 (0.782)	**0.54** (0.087)	0.54 (0.166)	0.52 (0.210)	0.54 (0.112)	0.54 (0.166)	0.52 (0.210)
24	**0.56** (0.014)	**0.54** (0.058)	0.51 (0.233)	0.55 (0.123)	0.54 (0.176)	**0.55** (0.081)	**0.56** (0.097)	0.54 (0.176)	**0.55** (0.086)

Note: Real-time, out-of-sample forecasts of the real price of crude oil, 1992M1–2021M1, versus the end-of-month no-change forecast. Regression-based direct forecasts using the NYSE Arca oil index. Brackets report the p-values for serial dependence robust tests of Pesaran and Timmermann (2009) for the null of no directional accuracy and of Diebold and Mariano (1995) for equal MSFEs relative to the end-of-month (traditional random walk) no-change forecast. Boldface values indicate statistically significant improvements over the end-of-month no-change forecast at the 10% significance level.

For all forecasts, parameters are estimated recursively using real-time data and forecasts are computed out-of-sample. We set $\alpha_h = 0$, which is common in practice and is found to provide better forecasts for all series, especially at the two-year horizon. Including a constant has only minimal effects on forecasts below one year.

The forecast performance is reported in Table 5. Columns 1–3 present the results for models estimated with monthly average oil prices. For all three oil series, the forecasts using monthly average prices fail to outperform the end-of-month no-change forecast at the one-month horizon.[13]

Columns 4–6 present the forecast performance of the disaggregated approach described by Eq. (14), while the last three columns present the performance of the PEPS approach, Eq. (15). These two alternative specifications perform remarkably similarly at all horizons and outperform the forecasts computed with monthly average oil prices at the one- to six-month horizons. Forecasts from these disaggregated approaches show MSFE ratios as low as 0.93 and success ratios as high as 0.58 at the one-month horizon. For Brent, the improvements in the MSFE and success ratio are also significant at the 5% significance level relative to the end-of-month no-change forecast at the one-month horizon. This finding is impressive, as it is the first time that any forecast of the monthly average real price of crude oil has been shown to yield significant improvements at this level for both MSFE and the success ratio at the one-month horizon relative to the end-of-month no-change forecast.

The MSFE gains are about 7% relative to the end-of-month no-change forecast. In terms of magnitude, the gains are more reminiscent of the short-horizon predictability occasionally documented for asset prices. The significant forecast gains for crude oil suggest some degree of predictability. However, the magnitude of the forecast gains relative to the traditional naive benchmark reinforces that the real price of crude oil is more similar to asset prices than previously concluded.

[13] The forecast gains relative to the monthly average no-change are very similar to those reported by Chen (2014) and are robust over time, as shown in the appendix Table A4 and Figure A2.

The regression estimates presented in Table 6 for Eqs. (13) and (15) provide useful intuition for our results. For the one-month-ahead forecast regressions using monthly average prices, the estimates of $\check{\beta}_1$ are around 0.6 and the R-squared is about 15%. On the contrary, the estimates of $\hat{\beta}_1$ and $\tilde{\beta}_1$ for the disaggregated regressions drop to 0.2–0.3, and the R-squares fall to 2%–4%. Despite the decrease in the coefficients and the measure of fit, the out-of-sample forecasts of Table 5 imply that the AMEX index provides some additional predictive power that is not contained in the last available daily oil price.

The intuition for these results becomes clearer when we consider the effect of averaging on the informational content of prices. Aggregation of daily prices into monthly averages results in a loss of information because it dilutes the latest information contained in the end-of-month price with stale information from previous days. In contrast, end-of-month prices are used for the AMEX index, which implies that this information loss does not occur for the predictor variable. As both the AMEX and oil prices are likely to respond to similar news about the global oil market, the use of the end-of-month AMEX index helps recover some information lost by averaging the price of crude oil. The results of this section suggest that most of the forecasting gains in the original specification of Chen (2014) stem from the recovery of lost information from averaging. However, the fact that we still find some improvement in the forecast accuracy for the disaggregated approaches indicates that the AMEX index contains some additional information beyond that contained in the last price of oil.

Our results indicate that even though the use of disaggregated oil prices is of first-order importance, the addition of external predictor variables can significantly improve the forecasts of the real price of crude oil. In the online appendix, we document similar gains for the RAC price series using the Commodity Research Bureau's (CRB) index of the price of industrial non-oil raw materials (used by Alquist et al., 2013; Baumeister & Kilian, 2012, among others). These findings are remarkable because it is the first time that any forecast of the monthly average real price of crude oil has been shown to significantly outperform the traditional random walk no-change forecast

Table 6
Coefficient estimates for one-month-ahead predictive regressions using the NYSE Arca (AMEX) Oil Index.

Sampling	Monthly average prices			End-of-month prices			Period-end-point sampling		
Dep. Variable	WTI	Brent	RAC	WTI	Brent	RAC	WTI	Brent	RAC
AMEX	0.593	0.614	0.548	0.237	0.193	0.196	0.309	0.313	0.257
	(0.067)	(0.069)	(0.063)	(0.053)	(0.060)	(0.050)	(0.080)	(0.090)	(0.075)
Constant	0.0001	0.0004	−0.0002	−0.001	−0.001	−0.001	0.003	0.005	0.003
	(0.004)	(0.004)	(0.004)	(0.003)	(0.003)	(0.003)	(0.005)	(0.006)	(0.005)
N	449	449	449	449	449	449	449	449	449
adj-R^2	0.148	0.149	0.142	0.040	0.020	0.032	0.030	0.024	0.024

Note: Coefficient estimates of $\hat{\alpha}$ and $\hat{\beta}$ from Eqs. (13)–(15) for different crude oil price series, 1992M1–2021M1. HAC standard errors in parentheses.

at the one-month horizon. We leave it to future research to evaluate whether other predictor variables that have been proposed will continue to contain predictive power against the traditional random walk naive forecast.

4. Robustness and potential extensions

The forecast gains from temporal aggregation are robust to a variety of alternative modeling choices. For example, nowcasting the CPI by extrapolating the last observed inflation rate, rather than the historical average inflation rate, has almost no effect on the forecasts themselves and does not change any of our conclusions. This robustness is consistent with previous findings (Baumeister & Kilian, 2012; Ellwanger & Snudden, 2023b) that fluctuations in the CPI deflator are small compared to fluctuations in nominal oil prices. The results are also unaffected by the choosing a later target vintage, as there are no revisions to the historical data post-2021M1. Finally, while the use of ex post revised data instead of real-time data can lead to minor differences in the forecast accuracy measures, the large forecast gains from employing disaggregated approaches remain qualitatively unchanged (see Appendix A2.5). At shorter forecast horizons, the differences arising from alternative data treatments are an order of magnitude smaller than the improvements achieved through disaggregated approaches, which highlights their first-order importance.

Accurate oil price forecasts have been shown to benefit policymakers by significantly aiding in projecting inflation expectations, industrial production, and producer price indexes, leading to welfare gains from improved policy responses (Degiannakis & Filis, 2023). Our results are directly applicable to their framework, which uses conditional projections of these variables based on oil price forecasts.

Our forecast evaluations focus on the two most common forecast criteria in the literature, MSFE and directional accuracy, for consistency with the previous literature. The gains from the disaggregated approaches are likely to extend to other forecast techniques, such as forecast averaging, and other evaluation criteria. For example, it can be shown that the improvements in the success ratio enhance the directional accuracy assessment of both upward and downward price movements. Moreover, forecasts that perform well in terms of MSFE often also contribute to better density forecasts. While properly examining forecasts along these dimensions is beyond the scope of this paper, using disaggregated oil price forecasts for alternative evaluation criteria offers a promising avenue for future research.

5. Conclusion

We showed how disaggregated approaches that exploit information from the underlying daily crude oil prices can be applied to model-based forecasts. Forecasts utilizing information in daily prices produced forecast improvements over the monthly average no-change forecast of up to 45% at the one-month horizon, which is unprecedented in this literature. The magnitude of the gains decreased with the forecast horizon, yet models estimated with daily prices still improved upon traditional forecasts as far ahead as the 24-month horizon.

We also directly tested, for the first time, whether model-based forecasts of the monthly real price of crude oil outperform the traditional random walk forecast, which is the end-of-month no-change forecast. We found that none of the current forecast practices estimated with monthly average prices outperformed the end-of-month no-change forecasts in terms of both the MSFE and directional accuracy at horizons lower than 24 months. By contrast, some disaggregated approaches displayed significantly lower MSFE ratios and improved directional accuracy relative to the end-of-period no-change forecast. In economic terms, the forecast gains relative to this benchmark are reminiscent of the short-horizon predictability documented for asset prices.

Incorporating nominal daily data into model-based forecasts for real period-average series presents a promising avenue for future research. Our study shows that even straightforward applications of disaggregated methods result in large forecast improvements, and thus highlights the value of daily data for forecasting average series.

CRediT authorship contribution statement

Amor Aniss Benmoussa: Conceptualization. **Reinhard Ellwanger:** Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **Stephen Snudden:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2025.02.009.

Data and code availability

Data and code availability The complete data and code replication package supporting the findings of this study are openly available in openICPSR at https://doi.org/10.3886/E222743V1, reference number openicpsr-222743. There are no restrictions on access or use of the data and code.

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